WORKED SOLUTION

STEP I 2005, Question 8

Pure • First Order Differential Equations

STEP I 2005, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s1-q8

The key idea

The opening identity is a recipe for spotting equations of the form $2xy\frac{dy}{dx} = ky^2 +$ (something in x alone). Once an equation is in that shape, the substitution $y^2 = x^k f(x)$ cancels the y^2 terms and leaves $f'(x)$ equal to a function of x, which we simply integrate. The only work is choosing k and, in part (ii), dividing through so that the left-hand side becomes $2xy\frac{dy}{dx}$.

Solution

The identity

Differentiate $y^2 = x^k f(x)$ implicitly with respect to x, using the product rule on the right:

$$2y\frac{dy}{dx} = kx^{k-1}f(x) + x^k\frac{df}{dx}.$$

Multiplying by x,

$$2xy\frac{dy}{dx} = k\,x^k f(x) + x^{k+1}\frac{df}{dx} = ky^2 + x^{k+1}\frac{df}{dx},$$

since $x^k f(x) = y^2$. This is the required result.

Part (i)

With $k = 1$ we write $y^2 = xf(x)$, and the identity says $2xy\frac{dy}{dx} = y^2 + x^2 f'(x)$. Comparing with the given equation, the y^2 terms match, so the equation holds exactly when

$$x^2 f'(x) = x^2 - 1, \qquad \text{that is,} \qquad f'(x) = 1 - \frac{1}{x^2} \quad (x \neq 0).$$

Integrating, $f(x) = x + \dfrac{1}{x} + c$, so

$$y^2 = xf(x) = x^2 + cx + 1.$$

The condition $y = 2$ when $x = 1$ gives $4 = 1 + c + 1$, so $c = 2$ and

$$y^2 = x^2 + 2x + 1 = (x+1)^2.$$

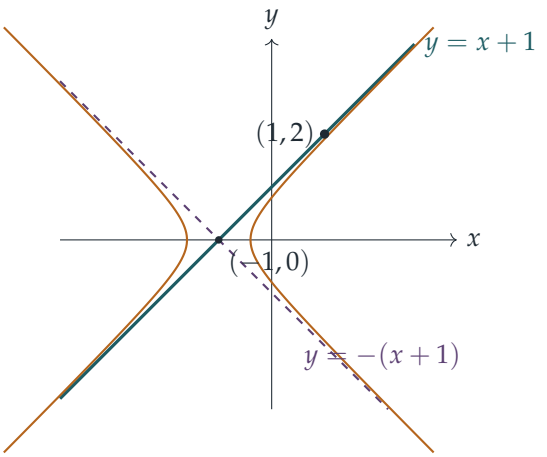
Hence $y = x + 1$ or $y = -(x+1)$; only $y = x + 1$ passes through $(1, 2)$, so the solution is

$$y = x + 1.$$

(Substituting back: $2x(x+1) \cdot 1 = 2x^2 + 2x$ and $(x+1)^2 + x^2 - 1 = 2x^2 + 2x$, so it satisfies the equation for every x, including $x = 0$ where we divided.)

Ask yourself What does "describe geometrically" want, when the answer is just $y = x + 1$?

The solution curve is a straight line: the line of gradient 1 through $(-1, 0)$ and $(0, 1)$, making an angle of $45°$ with the x-axis. It is worth saying why this is a little surprising. The general solution $y^2 - x^2 - cx = 1$ is a family of conics; completing the square gives $y^2 - (x + \frac{c}{2})^2 = 1 - \frac{c^2}{4}$, a rectangular hyperbola centred at $(-\frac{c}{2}, 0)$ with asymptotes $y = \pm(x + \frac{c}{2})$. The initial condition picks out $c = 2$, the case where the right-hand side is zero and the hyperbola degenerates into its own pair of asymptotes $y = \pm(x + 1)$. Our solution is the branch $y = x + 1$ of that pair.



The solid line is the solution; the dashed line is the other half of $y^2 = (x+1)^2$. The amber curves sketch the shape of the non-degenerate members of the family: a rectangular hyperbola approaching a pair of lines like these as asymptotes.

Part (ii)

To use the identity we need $2xy\,\frac{\mathrm{d}y}{\mathrm{d}x}$ on the left, so divide the equation by x (the $\ln x$ term forces $x > 0$, so this is safe):

$$2xy\frac{\mathrm{d}y}{\mathrm{d}x} = -y^2 + \frac{2\ln x}{x}.$$

The coefficient of y^2 is -1, so take $k = -1$, that is $y^2 = f(x)/x$. Then $x^{k+1} = x^0 = 1$ and the identity reads $2xy\,\frac{\mathrm{d}y}{\mathrm{d}x} = -y^2 + f'(x)$. Comparing,

$$f'(x) = \frac{2\ln x}{x}.$$

Since $\frac{\mathrm{d}}{\mathrm{d}x}(\ln x)^2 = \frac{2\ln x}{x}$, we get $f(x) = (\ln x)^2 + c$, and so

$$y^2 = \frac{(\ln x)^2 + c}{x}.$$

At $x = 1$, $y = 1$: $1 = (0 + c)/1$, so $c = 1$. Taking the positive square root, because $y = 1 > 0$ at $x = 1$ and y never vanishes (the numerator $1 + (\ln x)^2$ is at least 1), the solution is

$$y = \sqrt{\frac{1 + (\ln x)^2}{x}}, \qquad x > 0.$$

Checks

- In (i), the line $y = x + 1$ can be checked directly in the equation, as above; integrating the equation numerically from $(1, 2)$ (Runge–Kutta) gives $y = 3.000$ at $x = 2$ and $y = 6.000$ at $x = 5$.
- In (ii), $y^2 = (1 + (\ln x)^2)/x$ gives $y(3) \approx 0.8577$ and $y(0.2) \approx 4.2369$; a numerical integration of the differential equation from $(1, 1)$ agrees to many decimal places.
- The identity itself: try $k = 0$, where it collapses to $2xy\,y' = xf'$, which is just $(y^2)' = f'$ multiplied by x.

Common slips

- In (ii), using $k = 1$ out of habit. The sign of the y^2 term, after dividing by x, is what fixes $k = -1$.
- Forgetting to divide by x in (ii), and so trying to match $2x^2y\,y'$ with the identity.
- Leaving the answer to (i) as $y^2 = (x + 1)^2$ without choosing the branch through $(1, 2)$.

Beyond STEP

The substitution here is a disguised version of a standard trick: writing $u = y^2$ turns $2xy\,y' = ky^2 + g(x)$ into the *linear* equation $xu' - ku = g(x)$, and x^{-k} is its integrating factor. Equations that become linear after a power substitution are called Bernoulli equations, and they are among the first non-linear equations met in a university ODE course. Boyce and DiPrima, *Elementary Differential Equations*, treats them alongside integrating factors and exact equations.

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